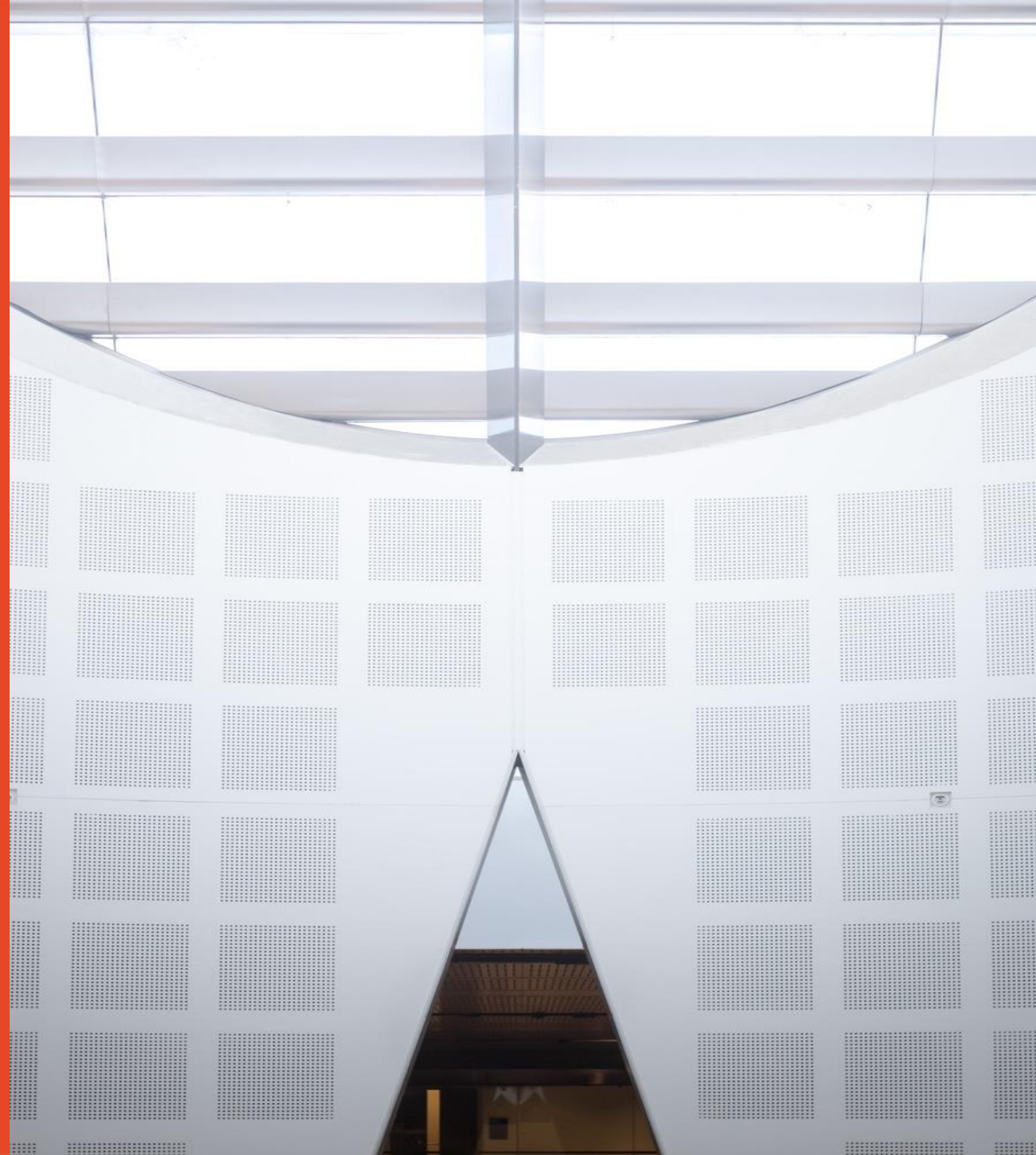


Solid State and Devices

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SYDNEY

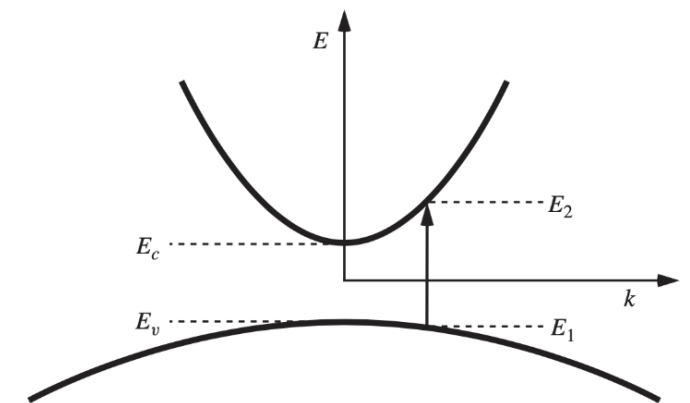
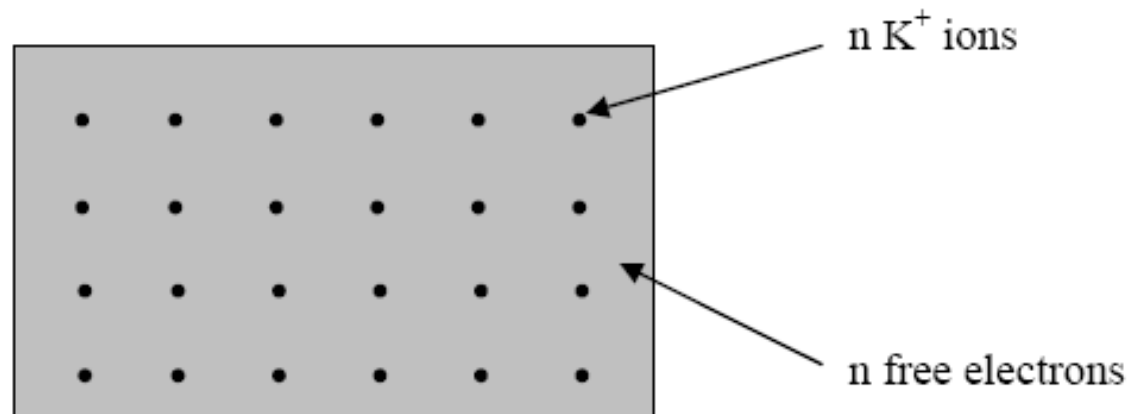


Density of States

is the number of available quantum states per unit energy that electrons can occupy in a material.
independent of whether those levels are filled with electrons or not.

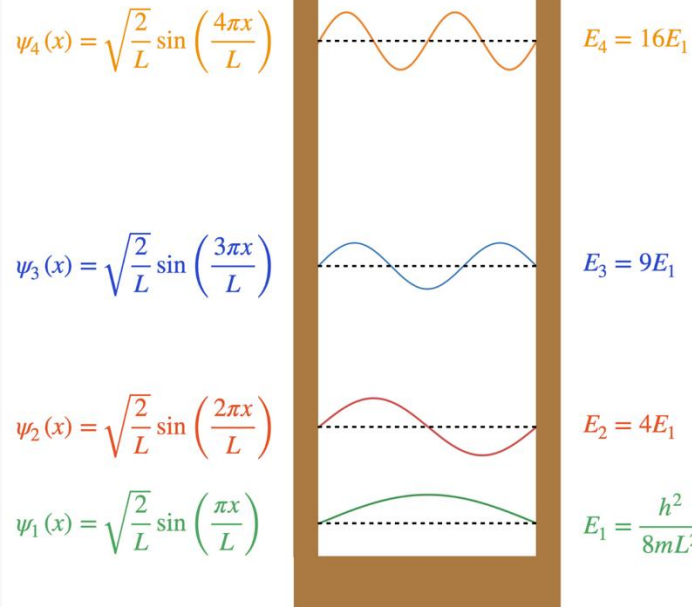
Electrons are treated as **free particles** moving in a constant potential (Good approximation near bands edge)

Allowed **k-states** are very dense (large crystal), so discrete states can be treated as **continuously distributed** in k-space.



Density of States

In 1D



to *quantise* the states and count them properly, we **imagine** the electrons are free, however in a cubic box

In 3D

In a cube of side length, $a=L$, the k space is quantised

Deriving the Total Number of Quantum States $N(E)$

$$\psi_{n_x, n_y, n_z}(x, y, z) = \sqrt{\frac{8}{a^3}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right)$$

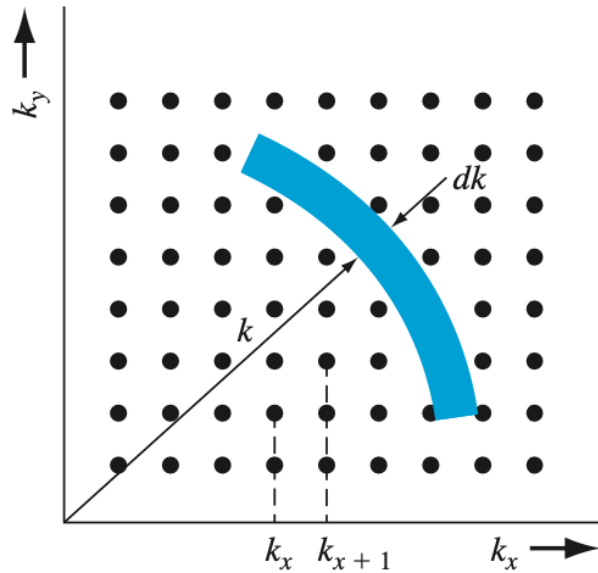
$$E_{n_x, n_y, n_z} = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2)$$

Introducing k

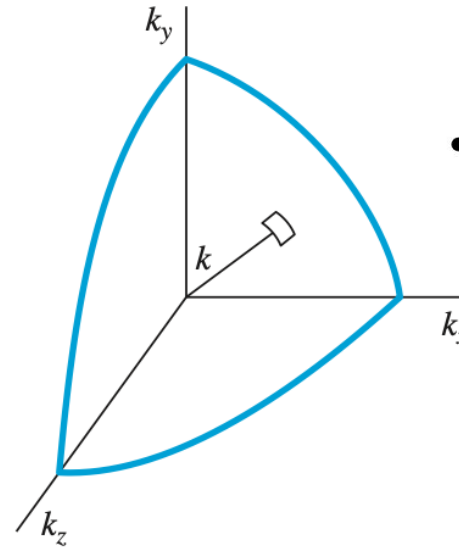
$$k_x = \frac{\pi n_x}{a}, \quad k_y = \frac{\pi n_y}{a}, \quad k_z = \frac{\pi n_z}{a}$$

$$E = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2) = \frac{\hbar^2 k^2}{2m}$$

Density of States



(a)



(b)

- From the boundary conditions of a particle in a box, the allowed wavevector components are quantized as:

$$k_x = \frac{\pi n_x}{a}, \quad k_y = \frac{\pi n_y}{a}, \quad k_z = \frac{\pi n_z}{a}$$

- The spacing between adjacent allowed values in any direction is:

In 1D

$$k_n = \frac{n\pi}{a}, \quad \frac{k}{n} = \frac{\pi}{a}, \quad \Delta k = \frac{\pi}{a}.$$

In 3D

- Therefore, each allowed state occupies a *cube in k-space* of volume:

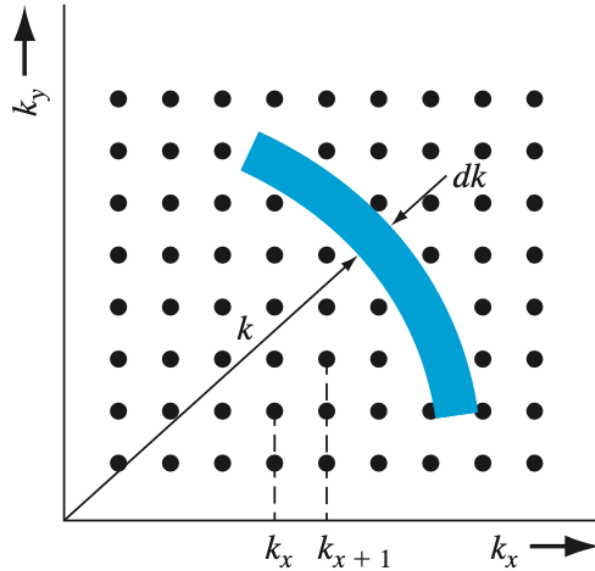
$$\Delta k^3 = \left(\frac{\pi}{a}\right)^3$$

- To find the number of discrete states inside a continuous volume of k -space (like a sphere), we divide that volume by Δk^3 , or equivalently, multiply by:

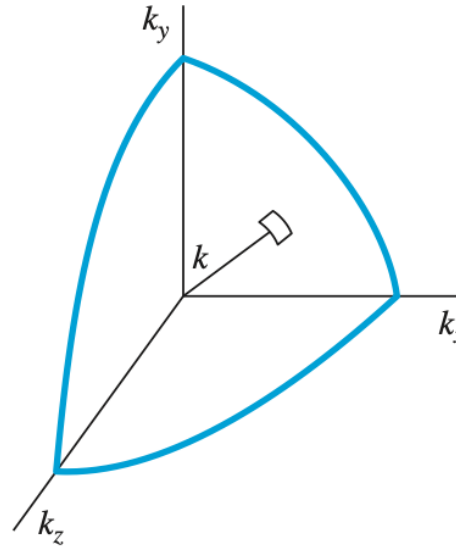
$$\left(\frac{a}{\pi}\right)^3$$

- This gives the ****density of allowed quantum states**** in k -space, and converts a volume in k -space into a count of states.

Density of States



(a)



(b)

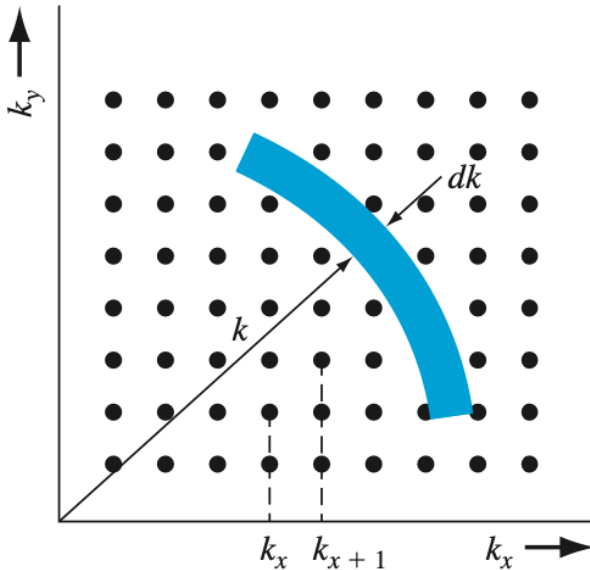
Lowest Energy Level

$$E = \frac{\hbar^2 k^2}{2m} \Rightarrow k = \sqrt{\frac{2mE}{\hbar^2}}$$

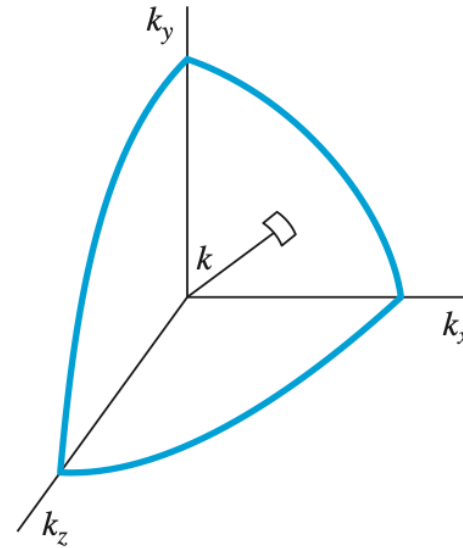
The number of quantum states with wavevectors inside a sphere of radius k in 3D k -space is given by:

$$N(E) = \underbrace{2}_{(1) \text{ Spin degeneracy}} \cdot \underbrace{\frac{1}{8}}_{(2) \text{ First octant only}} \cdot \underbrace{\frac{4}{3}\pi k^3}_{(3) \text{ Volume of sphere in } k\text{-space}} \cdot \underbrace{\left(\frac{a}{\pi}\right)^3}_{(4) \text{ Number of states per unit } k^3}$$

Density of States



(a)



(b)

total number of states per unit cell

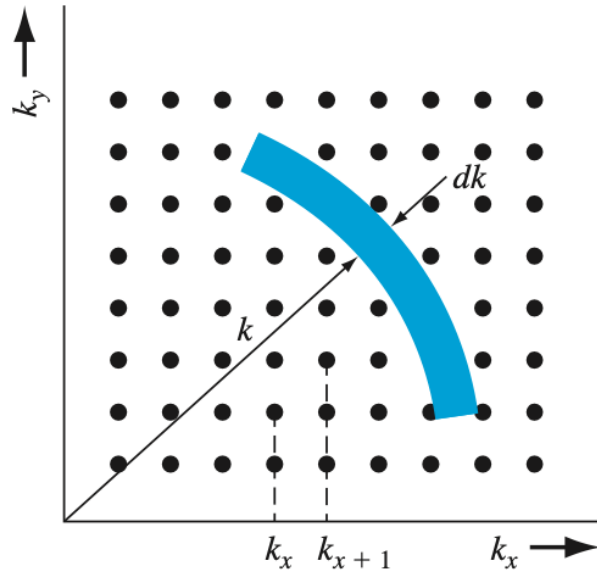
$$N(E) = \frac{a^3}{3\pi^2} \left(\frac{2mE}{\hbar^2} \right)^{3/2}$$

$$N(E) = 2 \cdot \frac{1}{8} \cdot \frac{4}{3} \pi k^3 \cdot \left(\frac{a}{\pi} \right)^3 = \frac{a^3}{3\pi^2} k^3$$

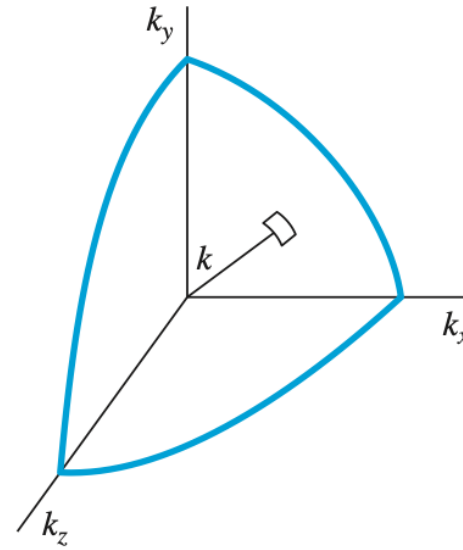
Note: The final result is now expressed in terms of k , but can also be written in terms of energy:

$$\text{Since } k = \sqrt{\frac{2mE}{\hbar^2}}, \Rightarrow N(E) = \frac{a^3}{3\pi^2} \left(\frac{2mE}{\hbar^2} \right)^{3/2}$$

Density of States



(a)



(b)

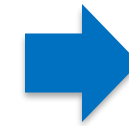
total number of states per unit cell

$$N(E) = \frac{a^3}{3\pi^2} \left(\frac{2mE}{\hbar^2} \right)^{3/2}$$

$$g(E) = \frac{dN(E)}{dE} = \frac{d}{dE} \left[\frac{a^3}{3\pi^2} \left(\frac{2mE}{\hbar^2} \right)^{3/2} \right]$$

$$g(E) = \frac{a^3}{3\pi^2} \cdot \frac{3}{2} \left(\frac{2m}{\hbar^2} \right)^{3/2} E^{1/2}$$

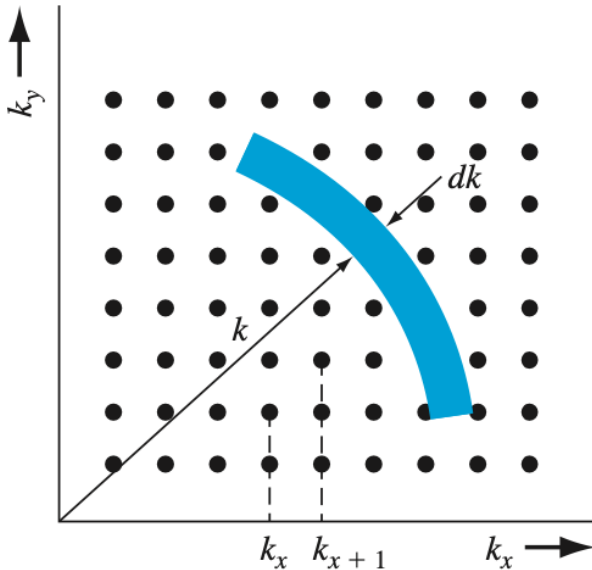
$$g(E) = \frac{a^3}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E}$$



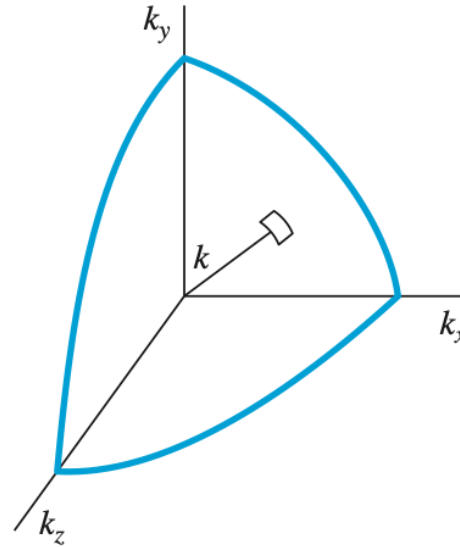
$$g(E) \propto \sqrt{E}$$

Density of States

In semiconductors not all states are available



(a)



(b)

$$g_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c} \quad \text{for } E \geq E_c$$

$$g_v(E) = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{3/2} \sqrt{E_v - E} \quad \text{for } E \leq E_v$$

$$E(k) = E_c + \frac{\hbar^2 k^2}{2m_e^*}$$

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*}$$

Valid for electrons and hole free to move

Limits and implications

$$g(E) = \frac{a^3}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E}$$

is valid under the following assumptions:

- **Parabolic Band Approximation:** The energy dispersion relation is assumed to be quadratic:

$$E(k) = \frac{\hbar^2 k^2}{2m^*}$$

This approximation holds only near the band extrema (i.e., conduction band minimum and valence band maximum).

- **Isotropic Bands:** The model assumes spherical constant-energy surfaces in k -space, which is valid for materials with isotropic effective mass.
- **Single Band:** Interactions between different bands are neglected. The formula applies to individual bands without mixing.
- **Non-degenerate Statistics:** The model assumes energies where the occupancy is low enough that Fermi-Dirac statistics reduce to the Maxwell-Boltzmann approximation (typically valid in lightly doped semiconductors at room temperature).

Limits and implications

Limitations

- **Invalid Far from Band Edges:** At high energies (far from band extrema), the band structure deviates from a simple parabola, making $g(E) \propto \sqrt{E}$ inaccurate.
- **Anisotropic Materials:** In materials with anisotropic effective mass (e.g., silicon), the true density of states differs due to ellipsoidal energy surfaces.
- **Low-dimensional Systems:** This model is strictly for 3D crystals. In 2D or 1D systems, the density of states behaves differently:
 - 2D: $g(E) = \text{const}$
 - 1D: $g(E) \propto \frac{1}{\sqrt{E}}$
- **Strongly Interacting Systems:** The model assumes independent particles; strong electron-electron interactions are neglected.

Limits and implications

Implications for Semiconductors

In semiconductors, the $\sqrt{E}$ dependence of $g(E)$ near the conduction band minimum (and similarly for holes in the valence band) leads to:

- A low density of states near the band edge: this is why thermal excitation at room temperature can populate only a small number of electrons into the conduction band unless doped.
- The number of available states increases rapidly with energy, which influences carrier concentration integrals:

The $\sqrt{E}$ form of $g(E)$ contributes significantly to the shape of this integral.

Limits and implications

Implications for Metals

In metals:

- The conduction band is partially filled even at 0 K, so states around the Fermi level are always occupied.
- The function $g(E) \propto \sqrt{E}$ still applies locally near the Fermi level if the band remains approximately parabolic.
- However, the actual DOS in metals is often computed numerically because the bands are complex, multi-valley, and partially filled over a wide energy range.

Conclusion: The $g(E) \propto \sqrt{E}$ model is a powerful and widely used approximation near band edges in semiconductors. While limited in scope, it underpins many essential semiconductor calculations including carrier statistics, intrinsic concentration, and Fermi level position.

To obtain the number of electron we need distribution function $f(E)$

$$n = \int_{E_c}^{\infty} g(E) f(E) dE$$

**Boltzmann distribution function
is not valid for electrons and holes at 0K**

Distribution Functions

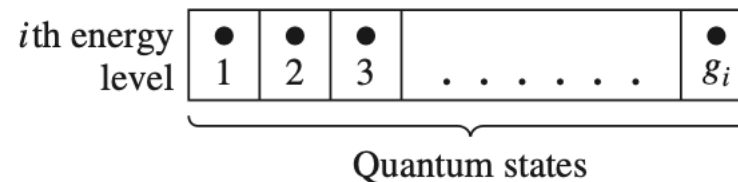
Band Structure indicates where electron can exist

Density of States indicates how many electrons can have a give E

We still have to obtain a function that quantify the distribution of electrons in the states

Then the total number of ways of arranging N_i particles in the i the energy level is

$$(g_i)(g_i - 1) \cdots (g_i - (N_i - 1)) = \frac{g_i!}{(g_i - N_i)!}$$



Therefore, the actual number of independent ways of realizing a distribution of N_i particles in the i th level is

$$W_i = \frac{g_i!}{N_i!(g_i - N_i)!}$$

Distribution Functions

For n energy levels, the indistinguishable total number of ways of arranging $(N_1, N_2, \dots, N_n)$ particles among n energy levels is the product of all distributions:

$$W = \prod_{i=1}^n \frac{g_i!}{N_i!(g_i - N_i)!}$$

In statistical mechanics, the **most probable distribution of particles** corresponds to the **maximum number of microstates**

$$\ln(\Omega) = \sum_i \ln \left(\binom{g_i}{n_i} \right) = \sum_i [\ln(g_i!) - \ln(n_i!) - \ln((g_i - n_i)!)] .$$

If we just tried to maximise $\ln W$ by differentiating with respect to N_i , we'd lose track of the fact that N and U must remain constant.

That's why we introduce **Lagrange multipliers**: they let us find the extremum of a function under constraints.

Distribution Functions

$$\ln(\Omega) = \sum_i \ln \left(\binom{g_i}{n_i} \right) = \sum_i [\ln(g_i!) - \ln(n_i!) - \ln((g_i - n_i)!)] .$$



Stirling approx.

$$\ln(n!) \approx n \ln(n) - n \text{ for large } n,$$



$$\delta \left[\ln(\Omega) - \alpha \sum_i n_i - \beta \sum_i n_i \varepsilon_i \right] = 0.$$

Considering also that:

$$\frac{\partial}{\partial n_i} \ln \Omega = \ln \left(\frac{g_i - n_i}{n_i} \right) .$$



Lagrange Multipliers

$$\mathcal{L} = \ln W - \alpha \left(\sum_i N_i - N \right) - \beta \left(\sum_i N_i E_i - U \right).$$

$$\frac{n_i}{g_i} = \frac{1}{\exp(\alpha + \beta \varepsilon_i) + 1}.$$

Define the Fermi level μ by $\alpha = -\beta\mu$ and let $\beta = 1/(k_B T)$, where T is the absolute temperature and k_B is Boltzmann's constant. Then:

$$f(\varepsilon_i) = \frac{n_i}{g_i} = \frac{1}{\exp \left(\frac{\varepsilon_i - \mu}{k_B T} \right) + 1}.$$

Distribution Functions

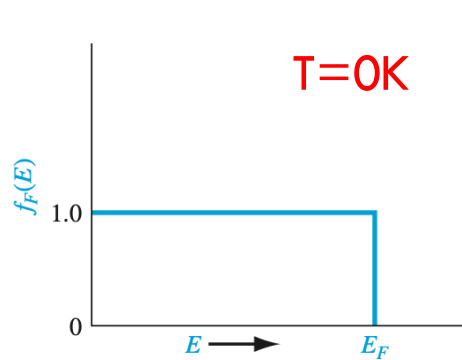


Figure 3.29 | The Fermi probability function versus energy for $T = 0$ K.

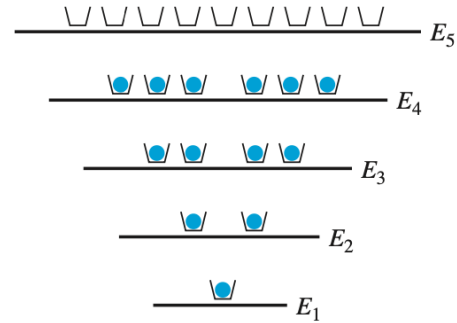


Figure 3.30 | Discrete energy states and quantum states for a particular system at $T = 0$ K.

$$f(E) = \frac{1}{\exp\left(\frac{E-\mu}{k_B T}\right) + 1}$$

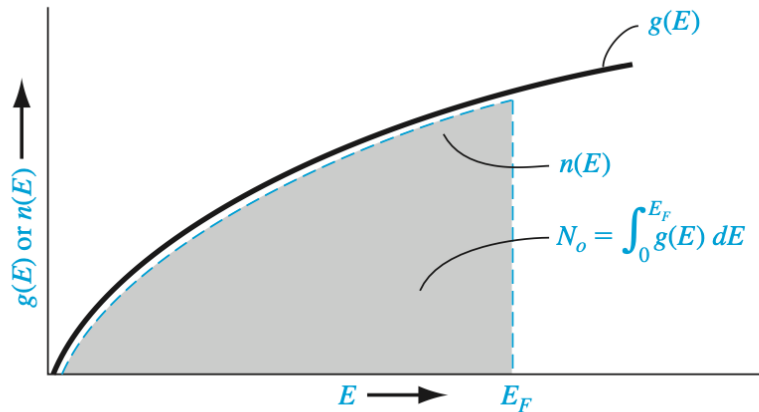


Figure 3.31 | Density of quantum states and electrons in a continuous energy system at $T = 0$ K.

At $T = 0$ K

- $f(E) = 1$ for $E < E_F$ (all states filled).
- $f(E) = 0$ for $E > E_F$ (all states empty).
- Step-function behaviour: the Fermi energy E_F marks the boundary between filled and empty states.
- Physical picture: electrons occupy the lowest possible energy states ("Fermi sea").

Distribution Functions

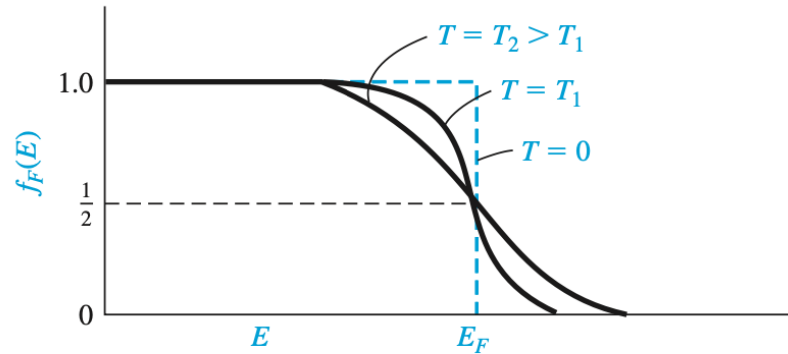


Figure 3.33 | The Fermi probability function versus energy for different temperatures.

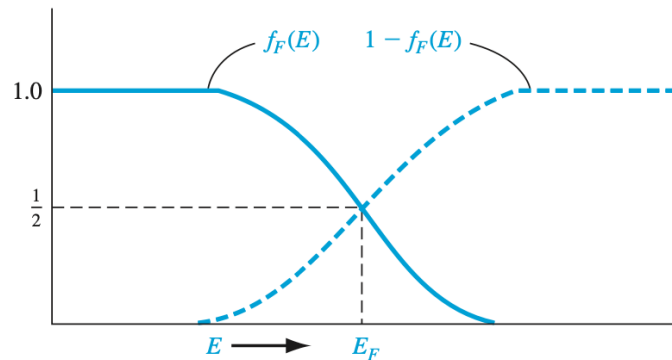


Figure 3.34 | The probability of a state being occupied, $f_F(E)$, and the probability of a state being empty, $1 - f_F(E)$.

$T > 0K$

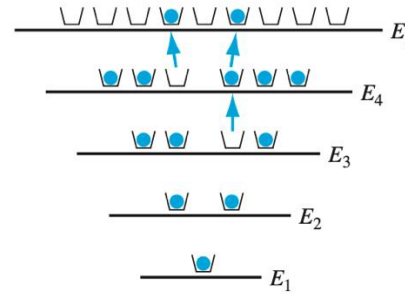


Figure 3.32 | Discrete energy states and quantum states for the same system shown in Figure 3.30 for $T > 0$ K.

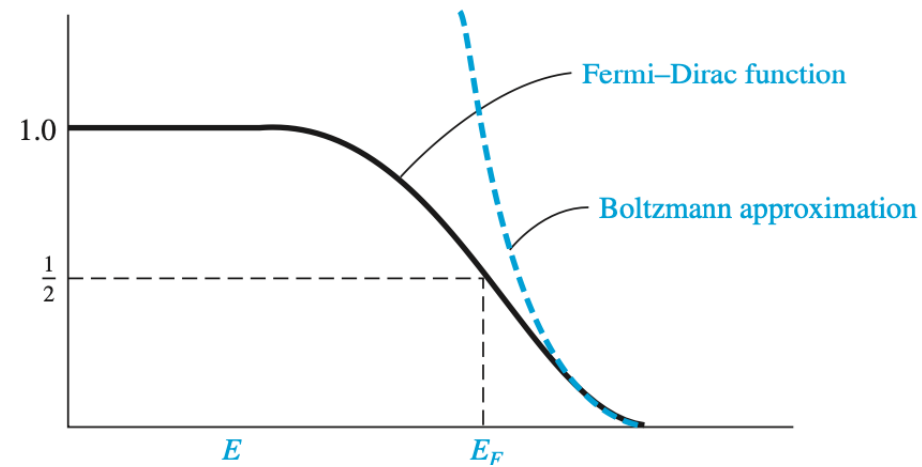
$$f(E) = \frac{1}{\exp\left(\frac{E - \mu}{k_B T}\right) + 1}$$

At $T > 0$ K (finite temperature)

- The step at E_F is "smeared out" over an energy range $\sim k_B T$.
- $f(E_F) = \frac{1}{2}$ (always half-filled at the Fermi energy).
- States just below E_F have some probability to be unoccupied, states just above E_F can be thermally occupied.
- Thermal excitations redistribute electrons around the Fermi level but **do not shift E_F significantly** (in metals).

Distribution Functions

$T \gg 0K$



At very high T (Boltzmann limit)

- For $E - \mu \gg k_B T$, the exponential dominates:

$$f(E) \approx e^{-(E-\mu)/(k_B T)}.$$

- Fermi-Dirac statistics reduce to the **classical Maxwell-Boltzmann distribution**.
- Physical meaning: quantum effects (Pauli exclusion, degeneracy pressure) are negligible, electrons behave like a classical gas.

Bose Einstein Distribution

For bosons: • Particles are indistinguishable (like Fermions). •

But There is no restriction on the number of particles per quantum state.

If an energy level i has g_i available states and is occupied by n_i bosons, the number of ways of distributing them is:

$$W_i = \binom{n_i + g_i - 1}{n_i} = \frac{(n_i + g_i - 1)!}{n_i! (g_i - 1)!}.$$

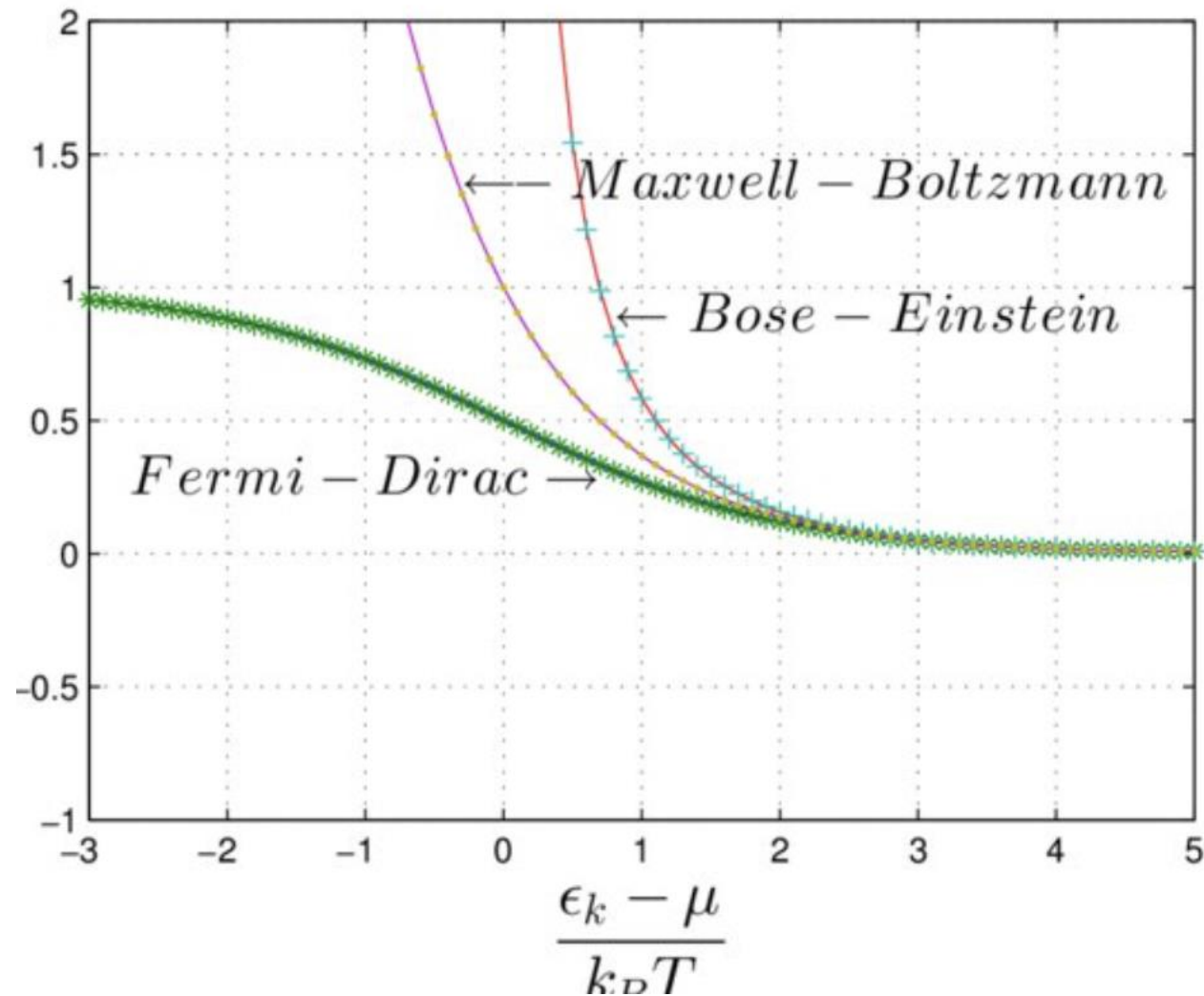
The total number of microstates is then

$$\Omega = \prod_i W_i = \prod_i \frac{(n_i + g_i - 1)!}{n_i! (g_i - 1)!}.$$

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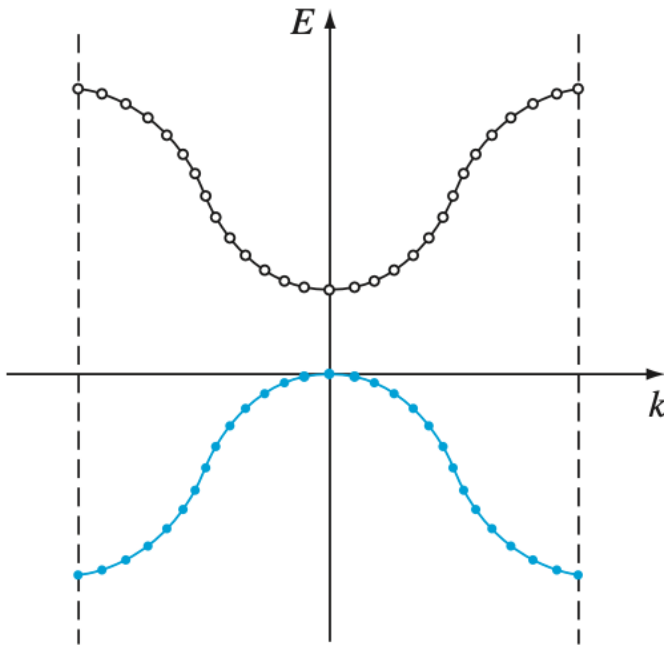
$$f_B(E) = \frac{1}{\exp\left(\frac{E - \mu}{k_B T}\right) - 1}.$$

Fermi-Dirac, Bose-Einstein, Boltzman Distribution



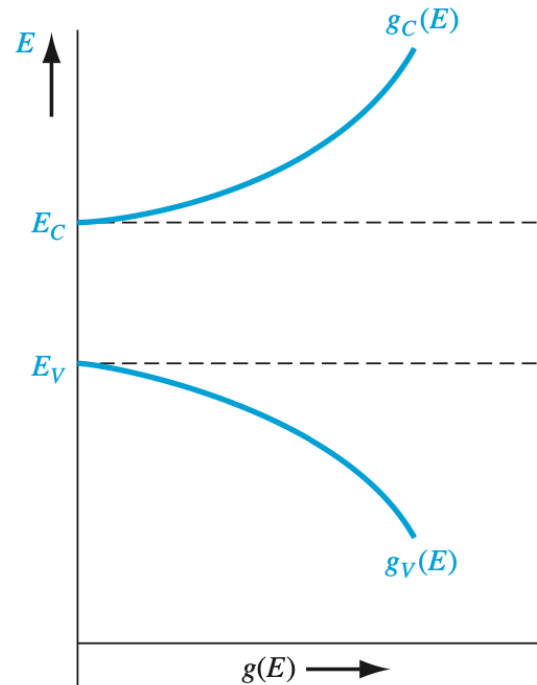
Band Structure, DOS and Fermi-Dirac distribution

Band Structure



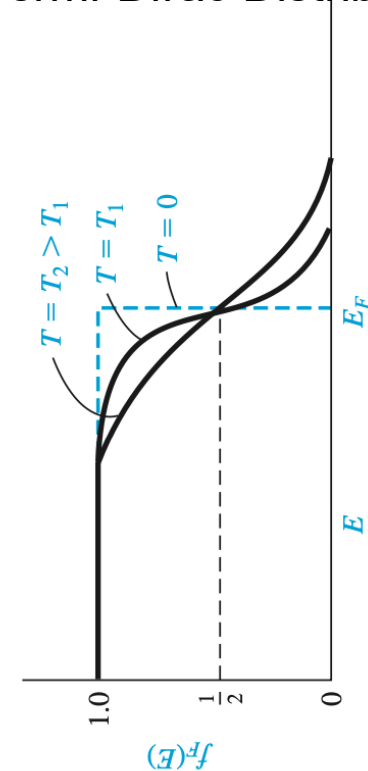
The **band structure** is nothing more than the set of allowed energy eigenvalues that come out when you solve the Schrödinger equation

DOS



Density of states counts how many states exist per unit energy and unit volume:

Fermi-Dirac Distribution



Fermi-Dirac indicates the occupancy of the states

Band Structure, DOS and Fermi-Dirac distribution (Real case)

